

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-2097

Pierce Brook 138 kV
July 03, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2097. The Interconnection Customer proposes a 1023.0 MW natural gas combined cycle facility interconnecting at the Pierce Brook 138 kV substation in the Pennsylvania area. The request was received on May 27, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 688 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$181,263,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
PIERCE BROOK – FARMERS VLY 138 KV	APEX – HAMMER 138 KV	550.5	187	100.0
APEX – HAMMER 138 KV	PIERCE BROOK – FARMERS VLY 138 KV	542.9	189	100.0
HOBE – FARMERS VALLEY 138 KV line	APEX – HAMMER 138 KV L	540.4	189	100.0
TENKILLER FERRY – LAS MONJAS 1416 138 KV	APEX – HAMMER 138 KV	382.8	188	54.4
LAY DAM – TENKILLER FERRY 138 KV	HAMMER – APEX 138 KV line	381.9	187	54.4
APEX 138-1 kV Auto	FARMERS VLY – PIERCE BROOK SUBSTATION 138 KV	207.7	354	64.8
MILES CITY – SIERRA 138 KV	APEX – HAMMER 138 KV	187.0	363	39.4
LAVA – SIERRA 138 KV	HAMMER – APEX 138 KV	186.5	365	39.4
LAVA – RODEO 138 KV	APEX – HAMMER 138 KV Line	185.9	368	39.4
RODEO – LAY DAM 138 KV line	APEX SUBSTATION – HAMMER 138 KV line	185.2	371	39.4
APEX 345/1kV Auto	FARMERS VALEY SUBSTATION – PIERCE BROOK 138 KV line	182.0	404	64.8
DPC BELL CENTER – JOHN KELLY 138 KV	TENKILLER FERRY SUBSTATION – LAS MONJAS 1416 138 KV	162.9	193	33.2
CACHE SLOUGH – MILES CITY 138 KV L	ANCHOR ENERGY – MILES CITY 138 KV	157.3	263	21.4

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
APEX – CACHE SLOUGH 138 KV line	ANCHOR ENERGY – MILES CITY SUBSTATION 138 KV	154.3	266	21.4
MILES CITY – ANCHOR ENERGY SUBSTATION 138 KV	APEX 138/1kV Transformer Bank	150.6	266	19.3
LAKE PLACID N – PONDEROSA 138 KV line	APEX 345-1 kV Transformer	144.9	184	21.5
PUNTLEDGE – SAN SIMEN-STEFANO 138 KV	APEX 138 kV/1 kV XFMR	132.8	195	5.6
LAY DAM 161/138 kV Auto	LAY DAM – RODEO 138 KV	129.1	130	20.7
PONDEROSA – LAKE PARK 138 KV line	APEX 345-1 kV Transformer Bank	127.7	214	21.5
LAKE PARK – DIVISION 138 KV line	APEX 345/1kV TX	122.8	226	21.5
NEW PLYMOUTH – PUNTLEDGE 138 KV	APEX 138/1kV Transformer Bank	121.6	222	5.6
APEX – DIVISION 138 KV	APEX 345/1kV TX	119.4	235	21.5
RED BLUFF – DPC BELL CENTER 138 KV line	TENKILLER FERRY – LAS MONJAS 1416 138 KV L	112.7	226	20.5
NE PLYMOUTH – AGUA FRIA 138 KV	APEX 345 kV/1 kV TX	104.9	227	5.0
RIVERMINE – ALSEA 138 KV L	LAS MONJAS 1416 – TENKILLER FERRY 138 KV L	102.6	213	27.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
PIERCE BROOK – FARMERS VLY 138 KV	Rebuild line with higher-capacity conductor	\$2,794,000	100% (\$2,794,000)
APEX – HAMMER 138 KV	Rebuild line with higher-capacity conductor	\$5,469,000	100% (\$5,469,000)
HOBE – FARMERS VALLEY 138 KV line	Rebuild line with higher-capacity conductor	\$5,240,000	100% (\$5,240,000)
TENKILLER FERRY – LAS MONJAS 1416 138 KV	Rebuild line with higher-capacity conductor	\$2,475,000	100% (\$2,475,000)
LAY DAM – TENKILLER FERRY 138 KV	Rebuild line with higher-capacity conductor	\$3,387,000	100% (\$3,387,000)
APEX 138 kV/1 kV TX	Replace transformer with higher-rated unit	\$5,642,000	100% (\$5,642,000)
SIERRA SUBSTATION – MILES CITY 138 KV L	Rebuild line with higher-capacity conductor	\$3,672,000	100% (\$3,672,000)
LAVA – SIERRA 138 KV	Rebuild line with higher-capacity conductor	\$7,424,000	100% (\$7,424,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
LAVA – RODEO 138 KV line	Rebuild line with higher-capacity conductor	\$1,718,000	100% (\$1,718,000)
RODEO – LAY DAM 138 KV line	Rebuild line with higher-capacity conductor	\$3,859,000	100% (\$3,859,000)
APEX 345/1 kV XFMR	Replace transformer with higher-rated unit	\$18,810,000	100% (\$18,810,000)
DPC BELL CTR – JOHN KELLY 138 KV	Rebuild line with higher-capacity conductor	\$8,086,000	100% (\$8,086,000)
CACHE SLOUGH – MILES CITY 138 KV L	Rebuild line with higher-capacity conductor	\$16,601,000	100% (\$16,601,000)
APEX – CACHE SLOUGH 138 KV line	Rebuild line with higher-capacity conductor	\$2,859,000	100% (\$2,859,000)
MILES CITY – ANCHOR ENERGY SUBSTATION 138 KV	Rebuild line with higher-capacity conductor	\$8,089,000	100% (\$8,089,000)
LAKE PLACID N – PONDEROSA 138 KV line	Rebuild line with higher-capacity conductor	\$8,354,000	100% (\$8,354,000)
PUNTLEDGE – SAN SIMEN-STEFANO 138 KV	Rebuild line with higher-capacity conductor	\$4,618,000	100% (\$4,618,000)
LAY DAM 161/138 kV Autotransformer	Replace transformer with higher-rated unit	\$5,284,000	100% (\$5,284,000)
PONDEROSA – LAKE PARK 138 KV line	Rebuild line with higher-capacity conductor	\$11,014,000	100% (\$11,014,000)
LAKE PARK – DIVISION 138 KV line	Rebuild line with higher-capacity conductor	\$10,990,000	100% (\$10,990,000)
NEW PLYMOUTH – PUNTLEDGE 138 KV	Rebuild line with higher-capacity conductor	\$29,682,000	100% (\$29,682,000)
DIVISION – APEX 138 KV line	Rebuild line with higher-capacity conductor	\$4,392,000	100% (\$4,392,000)
RED BLUFF – DPC BELL CTR 138 KV Line	Rebuild line with higher-capacity conductor	\$8,504,000	97% (\$8,262,000)
NEW PLYMOUTH – AGUA FRIA 138 KV	Reconductor line	\$817,000	100% (\$817,000)
RIVERMINE – ALSEA 138 KV	Reconductor line	\$1,725,000	100% (\$1,725,000)
	Total allocated to project		\$181,263,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.